An interpretable Dictionary Learning approach for Diagnostic Hypotheses from 3D Chest

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by

K.R.C.S. Kekulawala	-	EG/2021/4609
M.D.Y.B. Kularathne	-	EG/2021/4620
B.G.P.U. Thilakshana	-	EG/2021/4831
B.P.G.L. Thejana	-	EG/2021/4828

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Dr. Kaveen Liyanage	Mrs. Sithara Mahagama
(Supervisor)	(Co-Supervisor)

Abstract

Automated Diagnostic Hypotheses from 3D chest Computed Tomography requires clinically interpretable methods to support radiological decision-making. Current state-of-the-art (SotA) deep learning models often rely on post-hoc explainability techniques that provide approximate localizations and lack direct traceability to the models decisions. Furthermore, SotA models rarely support simultaneous multi-class classification and grounded localization, leading to a fragmented approach to diagnosis. This also hinders the ability to benchmark models across multiple dimensions of performance, including classification accuracy, segmentation quality, and explainability. This study aims to address these challenges through two contributions: First, we introduce an inherently interpretable dictionary learning framework in which detected abnormalities can be back-projected to the original voxel space, allowing diagnostic decisions to be traced back to relevant anatomical structures. Second, we develop a unified evaluation framework to benchmark SotA models across three axes: (i) classification accuracy, (ii) segmentation quality, and (iii) explainability, using standardized metrics and a publically available 3D chest CT dataset with expertly annotated abnormality localizations. Our method is benchmarked against SotA models using this framework, demonstrating competitive classification and segmentation performance while providing improved interpretability compared to deep learning baselines.

Acknowledgments

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We would like to acknowledge the creators and maintainers of the publicly available datasets used in this research, particularly the CT-RATE and RexGrounding-CT datasets and other medical imaging resources that have made this work possible. These datasets have been crucial for developing and validating the proposed methods.

We extend our thanks to the developers of the open-source frameworks and libraries used in this project, which significantly contributed to the implementation of this research.

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Acronyms

SotA	-	State-of-the-Art
CT	-	Computed Tomography
AI	-	Artificial Intelligence
VLFM	-	Vision Language Foundation Model
KSVD	-	K-Means Singular Value Decomposition
LC-KSVD	-	Label Consistent K-SVD
HU	-	Hounsfield Unit
DFDL	-	Discriminative Feature oriented Dictionary Learning

Chapter 1

Introduction

Developing a unified architectural pipeline for multi-class abnormality classification and grounded localization in 3D Computed Tomography (CT) addresses two of the most critical imperatives in clinical AI adoption: diagnostic accuracy and explainability. Specifically, utilizing dense pixel-level segmentation as the mechanism for localization forces the architecture to provide voxel-level visual evidence for its diagnostic claims. This ensures that the model is held accountable and its reasoning aligns with clinical diagnosis.

1.1 Background

The transition from 2D projectional radiography to volumetric CT represents a paradigm shift in medical image analysis.

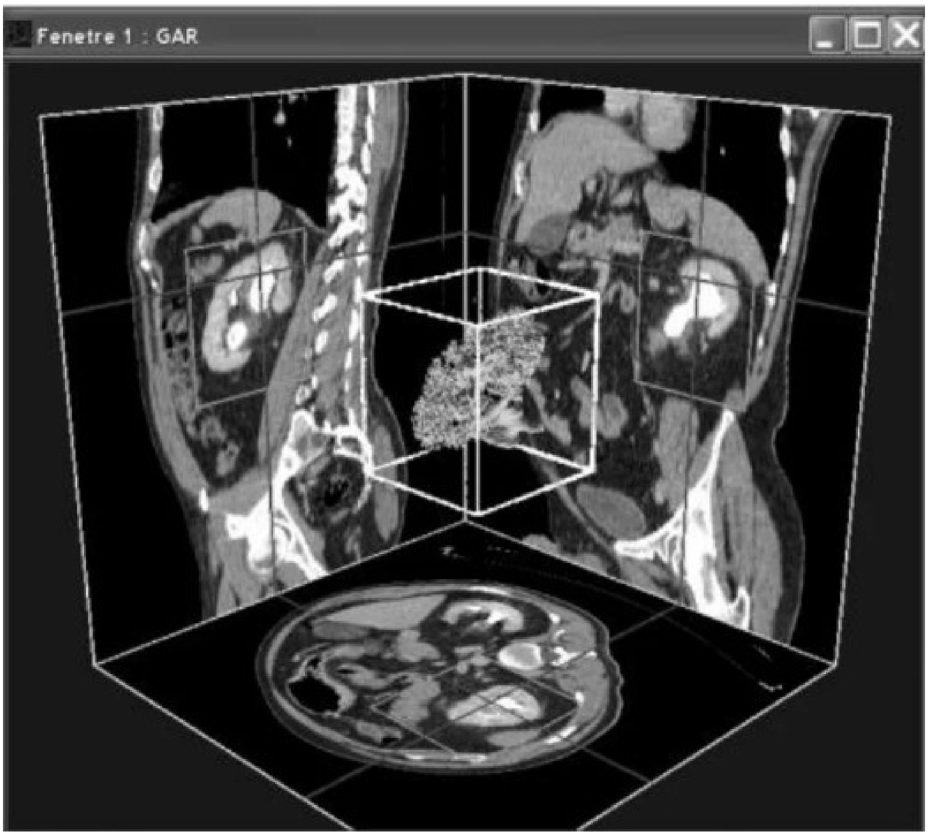

Figure 1.1: Transition from 2D projectional radiography to volumetric CT.

For Lung disease diagnosis, Chest X-rays (CXR) are commonly used as the first imaging test due to its low cost and low radiation exposure. However, studies show that their capability to detect abnormailities is limited when compared to 3D CT. [1].

Furthermore, CXR has historically benefited from the availability of massive, radiologist-annotated datasets, while the analysis of chest CT volumes has been severely constrained by a lack of such data. However, the recent introduction of large-scale, text-paired datasets has facilitated the development of state-of-the-art

(SotA) vision-language foundation models tailored specifically for volumetric imaging [2–4].

Beyond deep learning approaches, Dictionary Learning algorithms, particularly, Label Consistent KSVD (LC-KSVD) [5], Label Embedded Dictionary Learning (LEDL) and Fischer Discriminant Dictionary Learning (FDDL) offer viable approaches for interpretable CT analysis. All three techniques incorporate label information during the dictionary training process, and ensures that the sparse representation of the data is highly discriminative, making them well-suited for multi-abnormality tasks.

1.2 Problem Statement

The clinical utility of Deep Learning-based approaches is hindered by a lack of interpretability. While tools like Grad-CAM provide post-hoc explanations to model decisions, these are often insufficient for medical professionals who require a granular understanding of the model's decision making process. In high stakes environments, a model that produces a diagnosis without transparency in its reasoning is difficult to integrate into standard workflows, as it lacks the accountability necessary for legal and ethical compliance.

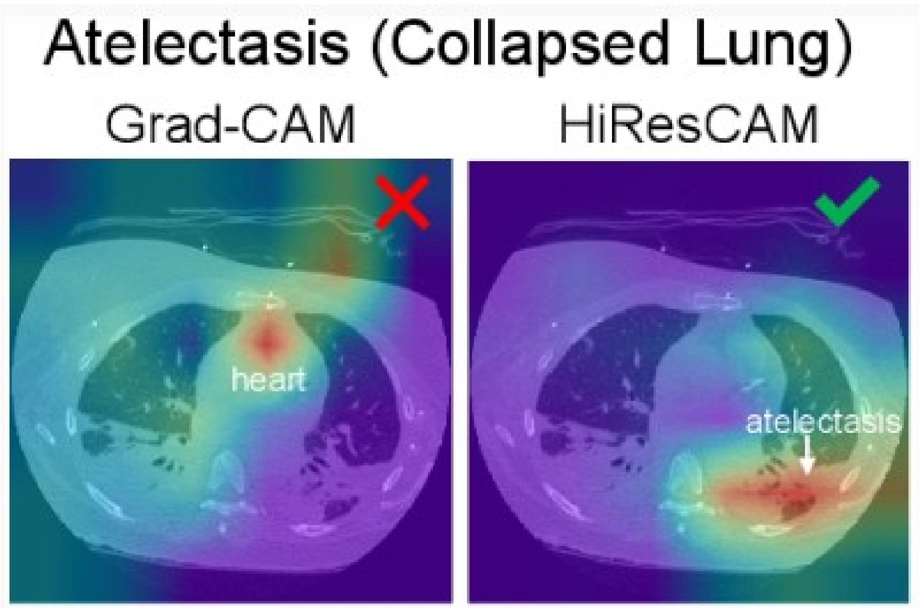

Figure 1.2: HiRes-CAM example illustrating higher-resolution localization compared with Grad-CAM.

1.3 Objectives and Scope

The primary goal of this project is to address the limitations of existing Deep Learning based approaches for Lung abnormality classification and localization by developing an inheerntly interpretable dictionary learning framework. This approach aims to enable transparent, anatomically grounded decision making, where each classification outcome can be traced back to specific pathological patterns. This aligns with clinical reasoning and facilitates validation and adoption in radiological workflows. The specific objectives are as follows:

1.3.1 Objectives

- Implement a Dictionary Learning approach to learn normal and pathological dictionary atoms using a publicly available annotated 3D chest CT dataset.
- Map dictionary atoms back to anatomical features and visualize their spatial contribution maps.
- Benchmark model performance against state-of-the-art classification and segmentation models on the same dataset.
- Evaluate abnormality localization by the model on a real-world dataset with expert radiologist review.

1.3.2 Scope

The scope of this study focuses on the development and evaluation of a dictionary learning based approach for classification and grounded localization of lung abnormalities using volumetric 3D CT data, with the following limitations:

- This study is limited to 3 thoracic abnormalities: Pulmonary Nodules, Ground-Glass Opacities, and Consolidations, based on expert radiologist recommendation and dataset availbility.
- This study will utilize a single publicly available dataset for training and evaluation, which may limit generalizability to other populations and scanner types.
- The proposed approach will be limited only to linear methods, and will not explore non-linear or hybrid models that provide better performance for less explainability.

1.4 Literature Review

The landscape of 3D CT analysis is currently divided into three primary architectural paradigms:

- Classification-based models that prioritize diagnostic labels
- Segmentation-based models that prioritize boundary precision
- Unified architectures designed for grounded localization

.

SotA models are predominantly specialized in either classification or segmentation. However, recent systems, particularly in 2025 and 2026, are moving toward multi-task learning or foundation models that can perform both diagnosis and voxel-level segmentation simultaneously.

Table 1 summarizes the key models in each category, their primary task orientation, and their performance on standard benchmarks.

1.4.1 Classification Architectures

Classification-based models for 3D CT have shifted away from 3D Convolutional Neural Networks (CNNs) such as 3D U-Net, 3D ResNet and DenseNet pipelines, toward Vision Transformers (ViTs) and self-supervised encoders. The benchmark for this paradigm is the CT-RATE dataset, which introduced the first public large-scale 3D medical imaging dataset with paired textual reports, enabling the development of Contrastive Language-Image Pretraining for Computed Tomography (CT-CLIP).

CT-CLIP is a VLFM designed for 3D chest CT analysis. It extends the CLIP framework to volumetric medical imaging by jointly learning representations from CT scans and their corresponding radiology reports. [6, 7] Through contrastive learning, the model aligns image features with textual descriptions, enabling tasks such as zero-shot classification, cross-modal retrieval, and report-guided abnormality localization. [6] The model is particularly effective in learning semantic relationships between radiological findings and volumetric imaging patterns such as GGO, consolidation, fibrosis, and pulmonary nodules. [6, 8]

Recent benchmarks such as Merlin and DINOv3 have further challenged the necessity of domain-specific pretraining. Merlin is a VLFM exclusively pre-trained on abdominal CT scans. Despite it's training, Merlin has consistently demonstrated exceptional zero-shot performance on chest CT classification tasks, even when it was restricted to a linear probe, significantly outperforming models like CT-CLIP, M3FM, and CT-FM that were explicitly trained on thoracic data, by an average of 12.3% in Area Under the Receiver Operating Characteristic Curve (AUROC). Specifically, in cross-domain retrieval tasks, Merlin surpassed CT-CLIP by 24.7%.

DINOv3, a self-supervised ViT pretrained on over 1.7 billion natural images, outperform specialized medical models like CT-Net and CT-CLIP across all metrics on several classification tasks due to it's ability to capture fine-grained textural features.

1.4.2 Segmentation Architectures

The prevailing SotA for supervised volumetric segmentation remains variations of the 3D U-Net, such as nnU-Net, which utilizes an adaptive framework to optimize preprocessing and architecture for specific datasets. However, the field has recently shifted toward foundation models capable of universal segmentation such as SegVol and SAM-Med3D. The SegVol model is the first 3D foundation model specifically designed for universal volumetric medical image segmentation. Similarly, SAM-Med3D adapts the Segment Anything Model (SAM) for 3D medical data.

Recent evaluations published at the Machine Learning for Healthcare (MLHC) Conference in 2025, which specifically utilized the ReXGroundingCT dataset, revealed that the current SotA text-prompted segmentation models that demonstrate near-perfect performance in general computer vision tasks struggle with the nuanced clinical descriptions and ill-defined boundaries of chest CT abnormalities.

1.4.3 Unified Architectures

Recent models have been designed to integrate abnormality detection, segmentation, and multimodal chain-of-thought reasoning, enabling pixel-level localization, structured report generation, and physician-like diagnostic inference in a single framework. BiomedParse, and it's cloud-deployed version MedImageParse, developed by Microsoft, are foundation models designed to parse 3D medical images. Other notable models include Citrus-V, MTMed3D and MedVista3D, where each architecture incorporates a multi-task learning (MTL) framework.

Table 1

1.4.4 Dictionary Learning Approaches

Applications of sparse dictionary learning in medical imaging have demonstrated significant success across diverse diagnostic tasks. In histopathological analysis, Discriminative Feature oriented Dictionary Learning (DFDL) has shown the ability to learn class specific dictionaries that achieve high classification accuracy even with limited training data a critical advantage when generous training samples are not available[19]. In neuroimaging, multi feature kernel supervised dictionary learning approaches have demonstrated superior performance in distinguishing Alzheimerś disease patients from cognitively unimpaired individuals, achieving 98.18% classification accuracy while maintaining fast testing times[20].

Most relevant to our work, dictionary learning has been successfully applied to lung disease diagnosis from CT images. One approach utilized both frozen dictionary learning and label consistent LC-KSVD to classify COVID 19 CT scans, achieving 89% accuracy on the CC-CCII dataset[21]. The frameworkś key advantage lies in its interpretability, where the sparse coding output can be directly mapped back to the original input signal domain. This work demonstrates that sparse representation methods offer an effective balance between performance and explainability for lung disease classification tasks, providing clear feature representations while maintaining competitive accuracy. Similarly, sparse representation methods have achieved 95.4% accuracy in classifying diffuse lung disease patterns on HRCT images, effectively handling the high variety and complexity of patterns[22].

1.4.5 Explainability Approaches

Grad-CAM is one of the most widely used post-hoc explanation methods for CNN-based classifiers [9]. It generates class-discriminative heatmaps by using gradients flowing into the final convolutional layer, allowing the user to identify image regions that influenced a prediction [9]. It's main limitation is that it produces low-resolution saliency maps and does not directly provide precise lesion boundaries or voxel-level localization, which is a major drawback for 3D CT interpretation [9**?**].

Seg-Grad-CAM extends this idea by adapting Grad-CAM to semantic segmentation outputs [**?**]. Instead of only highlighting regions that are important for image-level classification, it produces pixel-level relevance heatmaps for segmentation predictions [?]. This is particularly useful for volumetric medical imaging tasks where the target abnormality occupies a sparse region within a large 3D scan.

HiRes-CAM addresses one of the key weaknesses of Grad-CAM by producing higher-resolution and more faithful localization maps [?]. This is important in chest CT because many abnormalities, such as small nodules, consolidations, or subtle interstitial changes, may be missed by coarse heatmaps. HiRes-CAM aims to preserve fine spatial detail, making it more suitable for medical imaging applications where precise localization is required [?]. However, like other gradient-based methods, it remains a post-hoc explanation technique and does not change the internal decision process of the model.

1.5 Report Outline

This report is organized into five main chapters.

Chapter 1 introduces the research problem, background, objectives, scope, and literature review related to multi-class abnormality classification and grounded localization in 3D chest CT imaging. It also discusses existing deep learning and dictionary learning approaches together with their limitations in explainability.

Chapter 2 presents the theoretical background required for this study, including sparse representation, dictionary learning, K-SVD, LC-KSVD, frozen dictionary learning, and Vision Transformers used in medical imaging. In addition, important radiological abnormalities such as Ground Glass Opacity (GGO), consolidation, and pulmonary nodules are discussed.

Chapter 3 describes the proposed methodology of the project. This includes dataset acquisition, preprocessing steps, patch extraction, implementation of the LC-KSVD framework, abnormality classification pipeline, and the proposed back-projection mechanism used for interpretable localization and explainability generation.

Chapter 4 explains the experimentation and evaluation framework used in this research. It presents the selected baseline models, evaluation metrics, and the three evaluation axes: classification accuracy, grounded localization quality, and explainability. The chapter also summarizes the current progress of the project and planned evaluations.

Finally, Chapter 5 discusses the overall findings, contributions, limitations, and future improvements of the proposed interpretable dictionary learning framework for 3D chest CT abnormality analysis.

Chapter 2

Background

2.1 Dictionary Learning

2.1.1 Sparse Coding

TODO

2.1.2 KSVD

Dictionary learning is a signal modeling technique where a CT image patch is expressed as a linear combination of a small number of basis elements called dictionary atoms. [10] Instead of using all available features, the signal is approximated as:

$$y = Dx$$

where D represents the dictionary and x is a sparse coefficient vector containing mostly zero values. This produces a compact and interpretable representation. [11]

The dictionary D is typically learned using K-SVD (K-Singular Value Decomposition), which solves the following optimization problem:

$$\min_{D,X} \|P - DX\|_F^2 \quad \text{s.t.} \quad \forall i, \|x_i\|_0 \leq T$$

where P represents the training data, X is the sparse coefficient matrix, and T controls the sparsity level. K-SVD alternates between sparse coding and dictionary update steps to minimize reconstruction error. However, it does not incorporate class labels, which limits its discriminative ability for classification tasks. [10]

2.1.3 LC-KSVD

It jointly learns a dictionary, sparse codes, and a classifier so that the learned representations are not only good for reconstruction but also discriminative for classification through label consistency constraints. However, LC-KSVD still learns a single shared dictionary for all classes.

2.1.4 Frozen Dictionary Learning

To address this limitation, we adopt a two-stage dictionary learning strategy inspired by constrained dictionary learning approaches. [12] In the first stage, a dictionary is learned from normal CT scans to capture healthy anatomical structures. In the second stage, this dictionary is frozen and kept fixed, while additional atoms are learned only from abnormal scans. This approach can be interpreted as:

- preserving D_{normal} learned from healthy data
- learning D_{abnormal} independently for disease patterns

This separation ensures that normal anatomical structures are not overwritten, while abnormal features are explicitly isolated, improving interpretability and stability.

2.2 Radiological Patterns in Lung CT Imaging

2.2.1 CT Imaging

Computed Tomography (CT) imaging is widely used in lung disease diagnosis due to its ability to generate detailed cross-sectional images of lung structures. [13] CT scans provide high-resolution visualization of abnormalities such as Ground Glass Opacity (GGO), consolidation, Atelectasis, and pulmonary nodules, enabling accurate assessment and early detection of lung diseases. [14]

In modern medical imaging systems, CT imaging also supports computer-aided diagnosis and deep learning-based analysis for automated detection, segmentation, and classification of lung abnormalities. [15] These capabilities improve diagnostic efficiency and assist radiologists in clinical decision-making.

2.2.2 Ground Glass Opacity (GGO)

GGO appears as hazy regions in CT images which partially obscure underlying blood vessels or airway structures. Typically indicates partial filling of air spaces, inflammation, or early-stage infection. [16]

2.2.3 Consolidation

Consolidation appears denser than GGO, completely obscuring underlying lung structures such as vessels and airway walls. [14] It is often associated with severe infection, fluid accumulation, or advanced disease stages. The transition between GGO and consolidation is clinically important, as it can indicate disease progression or worsening lung conditions. [17]

2.2.4 Atelectasis

Atelectasis refers to the collapse or closure of alveoli in the lungs, leading to reduced lung volume and impaired gas exchange. It can be caused by various factors, including mucus plugging, foreign body aspiration, or compression from external masses. On CT imaging, atelectasis appears as areas of increased density with volume loss and may be associated with air bronchograms. [14]

2.2.5 Pulmonary Nodules

Pulmonary nodules are small, rounded lesions commonly observed in lung CT scans and are important indicators for early disease detection. They're usually less than 3 cm in diameter and can appear solid, subsolid, or ground-glass in nature. [18] Nodules are clinically significant as they may represent benign conditions such as infections or scars, but can also indicate early-stage lung cancer in suspicious cases. [19, 20] Careful monitoring of nodule size, shape, and growth over time is essential for accurate diagnosis.

Chapter 3

Methodology

We implement a six-stage pipeline for our study, consisting of:

1. Data Acquisition
2. Data Preprocessing
3. Implementation of the LC-KSVD Approach
4. Abnormality Classification
5. Back-Projecting Class-Discriminative Atoms and Deriving Explainability through Segmentation Masks
6. Evaluation and Benchmarking

3.1 Data Acquisition

Two public chest CT datasets: CT-RATE[21] and ReXGroundingCT[8] were used in our work. CT-RATE was used as the primary source of CT volumes (normal and abnormal). while ReXGroundingCT was used to obtain voxel-level segmentation masks per abnormality scan together abnormality labels, for each corresponding CT volume. Although CT-RATE also contains abnormality labels, RexGrounding-CT was chosen for its more clinically relevant set of findings, as well as its detailed voxel-level annotations that are essential for our explainability evaluation.

After an initial review of the thoracic findings available in ReXGroundingCT, a discussion with the supervising radiologist produced a shortlist of the following abnormalities that are clinically significant and have sufficient representation in the datasets:

1. Atelectasis
2. Lung nodule
3. Ground-Glass Opacity (GGO)
4. Consolidation

Both the volumes and their segmentation masks are formatted as NIfTI files. The masks are 3D volumetric segmentations ($H \times W \times D$) that align with the corresponding 3D CT scan, with their shape generally aligned with the scan's dimensions, as [512, 512, Depth]. For scans with multiple findings, masks can be structured as [F, H, W, D], where F represents the number of findings. Within a mask file, pixels are labeled with integer values corresponding to different entities for that finding, while 0 represents background.

3.2 Data Preprocessing

The data preprocessing pipeline consists of four major steps:

1. Resampling to isotropic spacing
2. Intensity windowing
3. Normalization
4. Patch extraction and vectorization

Each volume is resampled to 1.5 mm isotropic using trilinear interpolation to standardize the spatial resolution across scans. If the current spacing is already within 1% of target, this step is skipped. Each mask is also resampled to exactly match the resampled volume's spatial shape using nearest-neighbour interpolation to preserve integer label values

In the second step, a standard window Hounsefield unit (HU) of [-1000, +200] was used to separate the lung window from the CT volume. Then, a linear rescaling to [0, 1] as float32 was applied to the windowed volume, to normalize the intensity values.

Finally, the preprocessed volumes are divided into overlapping 3D patches of size $32 \times 32 \times 32$ with 32,768 features per patch (48 mm^3 at 1.5 mm spacing). Normal and abnormal scans are processed in separate phases to ensure that the Dictionary Learning model first learns a representation of normal anatomy before being exposed to abnormalities.

In phase 1, a maximum number of patches are sampled from each normal scan from any in-bounds location in the volume. all labelled class index 0

In phase 2, using the abnormal scans, we build a per-category binary mask by OR-ing all finding slots of the same category together; build a union mask across all categories; sample patch centres uniformly from the union mask foreground; each candidate patch must have $\geq$ 5% overlap with at least one category mask; the label is assigned to whichever category mask has the highest overlap with that patch (majority-class rule), which handles overlapping findings correctly

Both phases share the same maximum patches/scan budget; class balance is therefore determined by the normal/abnormal scan counts in the dataset, not a hard ratio All patches are assembled into X (32768 x N, float64) and H (N,) int64, then saved as .npz files.

3.3 Implementation of the LC-KSVD Approach

To implement the Dictionary Learning approach for our study, we required a concrete implementation of the LC-KSVD algorithm. [5] Currently, many of the available standard implementations of KSVD and LC-KSVD are developed using MATLAB, which is not ideal for integration with our Python-based data processing and evaluation pipeline. Although A few open-source Python implementations LC-KSVD are available, they either lacked an importable library, or did not provide the flexibility required for our study.

Therefore, we developed a custom implementation of the LC-KSVD algorithm in Python and published it on PyPI, the official third-party software repository for Python, under the name "reppi." The implementation follows a modular design, consisting of four separate components:

1. SRC - Utility Functions shared across Sparse Coding Algorithms

2. OMP - Implementation of Batch-OMP and OMP-Cholesky approaches for Sparse Coding

3. KSVD - Implementation of KSVD Dictionary Learning Algorithm

4. LC-KSVD - Implementation of LC-KSVD1 and LC-KSVD2 Dictionary Learning Algorithms

This design also allows for flexibility in integrating alternative sparse coding methods in the future. The source code is available at github.com/ckekula/reppi.

This modular architecture enables straightforward integration of the algorithm through library imports, while providing the flexibility required to experiment with different hyperparameters and configurations. The implementations of the SRC, OMP, and LC-KSVD modules are provided in Appendix A, Appendix B, and Appendix C, respectively.

3.4 Model Training and Inference

Prior to training, each patch vector is L2-normalised to unit norm, ensuring that the sparse coding step operates on texture direction rather than absolute intensity magnitude. Patches with a norm below 1×10^{-10}, corresponding to uniform zero regions with no signal, are discarded entirely.

The model is trained using LC-KSVD2, a dictionary learning algorithm that jointly optimises three objectives: reconstruction accuracy, label consistency, and linear classification. Training proceeds in two stages: 20-iteration K-SVD warm-start that initialises the dictionary on reconstruction error alone, followed by 50 LC-KSVD2 iterations that refine it under the combined objective. This two-stage procedure avoids poor local minima that arise when the label-consistency and classifier terms are applied before the dictionary has converged to a geometrically meaningful initialisation.

The key hyperparameters are a dictionary size of K = 128 atoms, a sparsity constraint of T = 10 non-zero coefficients per patch (approximately 8% of K), a label-consistency weight of $\alpha = 4.0$, and a classifier weight of $\beta = 2.0$. On smaller datasets where the patch count N is insufficient to support 128 atoms, the dictionary size is adaptively capped at $max(8, N/2)$ and the sparsity constraint is scaled proportionally.

At inference time, a new CT volume is loaded and preprocessed through the identical pipeline used during training. However, unlike training, instead of randomly sampling patches as in training, inference requires spatial exhaustiveness so that every region of the volume receives a classification score.

This is achieved by extracting patches on a dense regular grid with a stride of half the patch width along each spatial axis. For a volume of size 512^3, this produces $32^3 = 32768$ patches, each a 32^3 voxel window centred on a grid point. The 50% overlap between adjacent patches ensures that every voxel is covered by up to 8 patch windows, providing multiple independent score estimates per spatial location for the subsequent back-projection step.

3.5 Back-Projecting Class-Discriminative Atoms

The unique intuition we propose in our approach is to generate spatial contribution maps through back-projection. Because each atom in the dictionary D is associated with a specific class label through the matrix Q [5], the model can theoretically "explain" its decision for a test patch by examining which atoms were used in its sparse representation.

To generate a spatial contribution map for a class c, the algorithm identifies all atoms d_k that have been assigned to that class. For a query voxel or patch with sparse coefficients x, the contribution C_c is calculated as the partial reconstruction:

$$C_c = \sum_{k \in \text{Class } c} d_k x_k$$

This process back-projects the class-specific features into the original voxel space, creating a map that highlights the regions contributing to the detection of a specific abnormality. for class-discriminative atoms operating on raw voxel-space inputs, the back-projection is an exact, closed-form spatial attribution.

3.6 Deriving Explainability through Segmentation Masks

To evaluate the clinical accuracy of LC-KSVD, these contribution maps are converted into binary segmentation masks for comparison with ground truth annotations. This conversion can be achieved through various thresholding methods.

Grad-CAM's approximation has two sources of imprecision — the gradient itself is a local linear approximation of a non-linear function, and the spatial upsampling from the coarse feature map to input resolution introduces further interpolation error. LC-KSVD's contribution map has only one source of imprecision: the thresholding step you apply to binarize it into a mask. The map itself — before thresholding — is exact given the sparse code and the dictionary.

Unlike the passive Grad-CAM approach, atom back-projection is generative. The atoms are what the model actually used to reconstruct and classify the signal. There's no proxy involved. The localization evidence is produced by the same computation that produced the classification, not by a separate post-hoc analysis run afterwards.

Chapter 4

Evaluation Framework and Results

4.1 Evaluation Framework Design

Comparing a sparse linear approach with state-of-the-art non-linear model architectures using only raw performance scores can give an incomplete view of model behavior. In medical imaging, a useful model should not only predict the correct abnormality, but also identify the relevant anatomical region and provide a meaningful explanation for its decision. Therefore, the evaluation framework in this study is organized around three complementary evaluation axes:

- **Axis A: Classification Accuracy** evaluates the ability of the model to detect the presence or absence of each abnormality using standard discriminative metrics such as AUROC, precision, recall, and F1 score.
- **Axis B: Grounded Localization Quality** evaluates whether the model localizes the abnormality in the correct region of the 3D CT volume using spatial overlap and hit-based localization metrics such as DSC, IoU, and HIT@K.
- **Axis C: Interpretability and Explanation Faithfulness** evaluates whether the model's explanation or attribution map is spatially aligned with the true abnormality region using metrics such as Attribution-Mask IoU, Pointing Game Accuracy, and Energy Inside Mask.

Axis C is particularly important for this study because the proposed LC-KSVD-based approach is designed to provide interpretable decision-making through dictionary atoms and sparse activations. In contrast, most deep learning and foundation models require post-hoc attribution techniques such as Grad-CAM, HiResCAM, or saliency-based methods to explain their predictions. These post-hoc methods provide useful approximations, but they do not directly represent the internal decision process of the model. Therefore, the evaluation framework explicitly separates predictive performance, localization quality, and explanation faithfulness to better highlight the strengths and limitations of each model type.

4.2 Evaluation Metrics

A common set of quantitative metrics was used to evaluate the models across the three axes. This ensures that different models are compared under the same evaluation

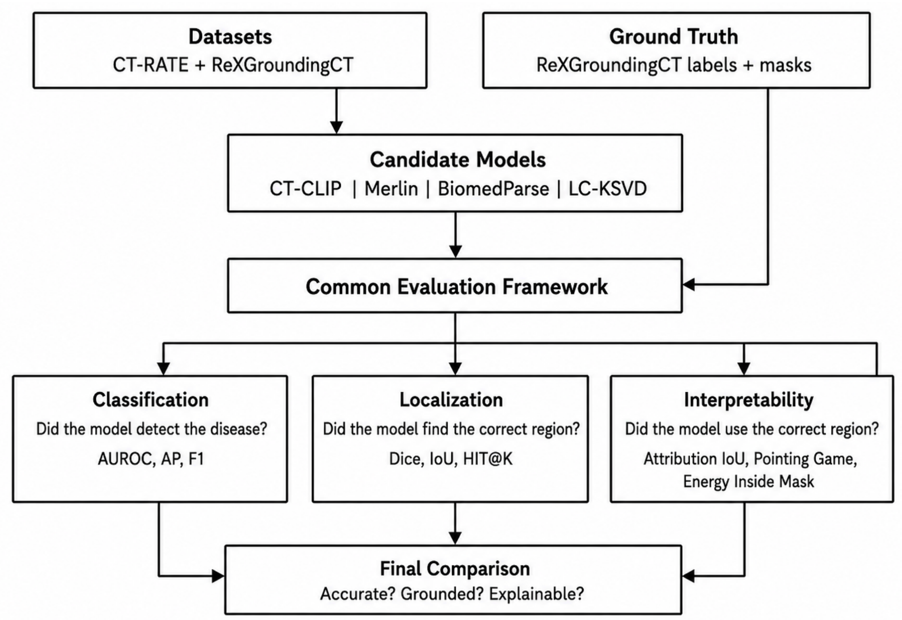

Figure 4.1: Architecture diagram of the proposed evaluation framework.

criteria, even though their raw outputs may differ. Classification-oriented models are evaluated using disease probability scores, localization-oriented models are evaluated using predicted masks, and interpretable models are additionally evaluated using attribution or activation maps.

4.2.1 Classification Metrics

For Axis A, the classification task measures whether the model correctly detects each abnormality. The following metrics are used:

- **Area Under the Receiver Operating Characteristic Curve (AUROC):** AUROC measures how well the model separates positive and negative cases across different decision thresholds. It is useful for imbalanced medical datasets because it evaluates ranking performance rather than relying on a single fixed threshold.

- **Precision and Recall:** Precision measures how many predicted positive cases are actually positive, while recall measures how many true positive cases are correctly detected by the model.

- **F1 Score:** F1 score is the harmonic mean of precision and recall. It provides a balanced measure when both false positives and false negatives are important.

- **Average Precision:** Average precision summarizes the precision-recall curve and is useful when positive cases are relatively rare.

4.2.2 Localization and Grounding Metrics

For Axis B, the localization task measures whether the predicted abnormality region overlaps with the ground-truth lesion mask. This is important because a model may correctly predict the presence of a disease while still focusing on the wrong anatomical area.

- **Dice Similarity Coefficient (DSC):** Dice measures the spatial overlap between the predicted mask and the ground-truth mask. It is defined as:

$$DSC = \frac{2|X \cap Y|}{|X| + |Y|}$$

where X is the predicted mask and Y is the ground-truth mask.

- **Intersection over Union (IoU):** IoU, also known as the Jaccard index, measures the ratio between the overlapping region and the total combined region:

$$IoU = \frac{|X \cap Y|}{|X \cup Y|}$$

- **HIT@K:** HIT@K measures the proportion of cases where the model achieves at least a small amount of meaningful overlap with the ground-truth lesion region. In this work, HIT@5, HIT@10, and HIT@25 are used to indicate whether the Dice score exceeds 0.05, 0.10, and 0.25 respectively. This is useful for diffuse abnormalities where exact boundary agreement is difficult.

4.2.3 Interpretability Metrics

For Axis C, the interpretability task evaluates whether the model's explanation is focused on the clinically relevant abnormality region. This is different from localization because the goal is not only to check where the model predicts a lesion, but also to determine whether the model's decision is supported by the correct anatomical evidence.

- **Attribution-Mask IoU:** This measures the overlap between the thresholded attribution map and the ground-truth lesion mask. A higher value indicates that the explanation is spatially aligned with the true abnormality.
- **Pointing Game Accuracy:** This checks whether the highest-attribution point lies inside the ground-truth lesion region. If the maximum response is inside the lesion mask, the case is counted as a hit.
- **Energy Inside Mask:** This measures the percentage of total attribution energy that lies inside the ground-truth mask. A high value indicates that the model's explanation is concentrated on the true abnormality region rather than irrelevant background structures.

4.3 Classification Accuracy Evaluation

The classification performance was evaluated across four abnormality classes: Lung Nodule, Lung Opacity, Consolidation, and Atelectasis. The proposed LC-KSVD-based baseline was compared with three foundation model baselines: Merlin, CT-CLIP, and BiomedParse. For LC-KSVD, consolidation-related performance was also used as a proxy reference for atelectasis due to current dataset and label constraints.

4.3.1 Overall Classification Performance

Figure 4.2 shows the macro-level classification performance across the evaluated models. Merlin achieved the strongest discriminative performance, with a macro-AUROC of 0.966. In contrast, CT-CLIP and BiomedParse produced lower macro-AUROC values of 0.538 and 0.509 respectively on the ReXGroundingCT evaluation setting.

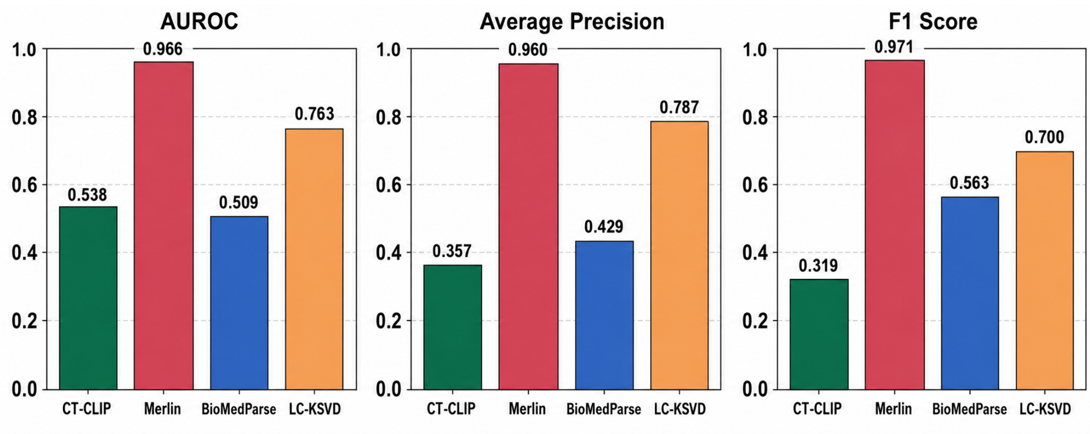

Figure 4.2: Overall classification performance comparison across models.

These results suggest that Merlin is the strongest classification baseline in this setup. CT-CLIP and BiomedParse show weaker classification performance, which may be due to differences in training objectives, dataset alignment, and the use of zero-shot or proxy classification outputs. BiomedParse is mainly designed for text-guided segmentation and grounding, so its classification score should be interpreted as a proxy rather than as the output of a dedicated classifier.

4.3.2 Per-Class Analysis

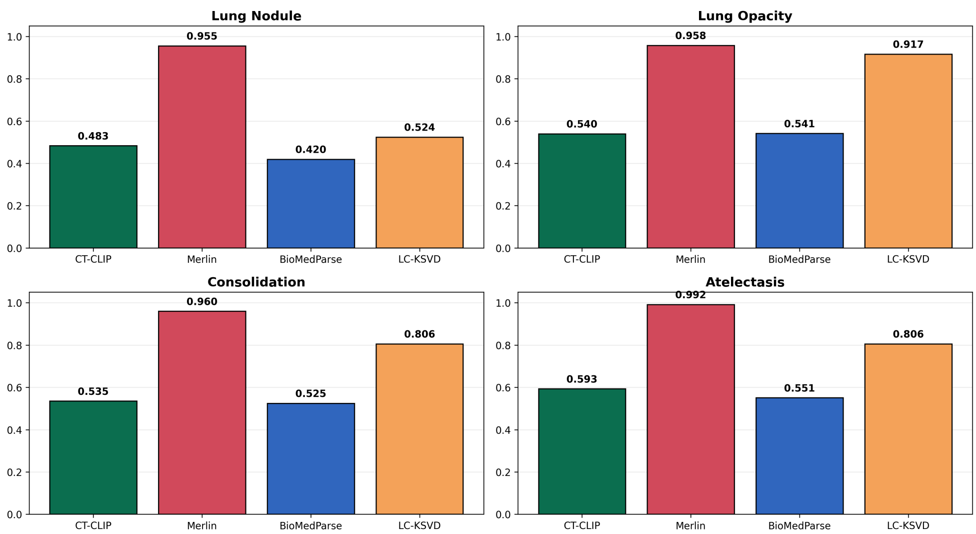

Figure 4.3: Per-class AUROC comparison Across Models.

The per-class AUROC comparison in Figure 4.3 provides a more detailed view of model behavior. Merlin maintains strong performance across all evaluated classes.

The LC-KSVD baseline shows promising results for non-focal or broader abnormalities, achieving AUROC values of 0.917 for Lung Opacity and 0.806 for Consolidation. It also achieves 0.806 for Atelectasis in the current evaluation setting. However, its performance for Lung Nodule is weaker, with an AUROC of 0.524.

This behavior suggests that LC-KSVD may capture broader abnormal tissue patterns more effectively than small focal lesions. Lung nodules are compact and localized, so their detection may require finer spatial resolution, multi-scale patch modeling, or more nodule-specific training samples. In contrast, abnormalities such as lung opacity and consolidation may produce more distributed visual patterns that are more compatible with the sparse dictionary representation.

4.4 Grounded Localization Evaluation

Axis B evaluates whether the models can identify the correct abnormal region in the volumetric CT scan. This is different from classification because a model can correctly detect the presence of a disease while still localizing the wrong anatomical region. The localization evaluation therefore uses Dice, IoU, and HIT@K metrics to measure spatial agreement between predicted lesion regions and ground-truth annotations.

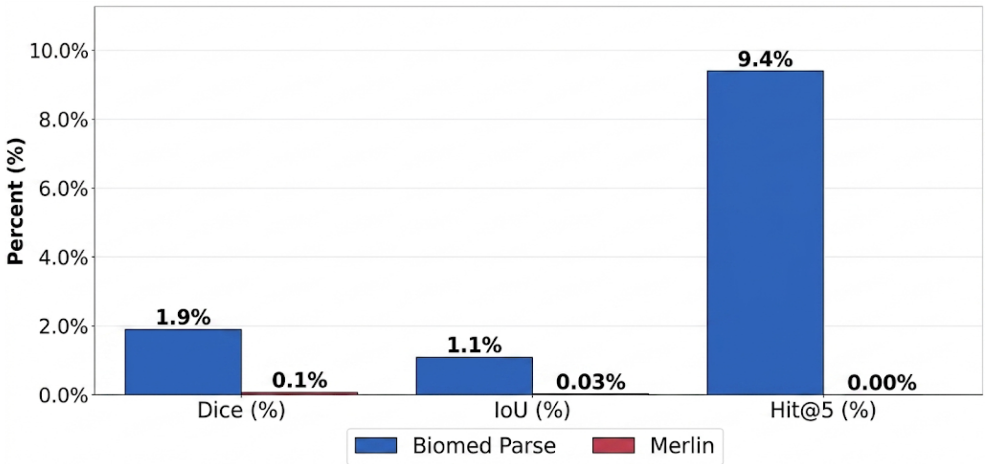

Figure 4.4: Overall localization performance comparison for BiomedParse and Merlin.

Figure 4.4 shows that grounded localization remains challenging for the evaluated models. The Dice, IoU, and HIT@K values indicate limited agreement between predicted lesion regions and reference annotations. This highlights an important gap between disease recognition and true spatial grounding. For clinical support, it is not sufficient to detect an abnormality; the model should also identify the correct location and extent of the abnormal region.

4.4.1 Per-Class Localization Performance

Figure 4.5 shows the per-class Dice comparison. The results indicate that localization performance varies substantially across lesion categories. This variation is likely influenced by lesion morphology, size, contrast, and boundary clarity.

Focal lesions such as lung nodules show comparatively better localization than diffuse abnormalities such as lung opacity, consolidation, and atelectasis. This suggests

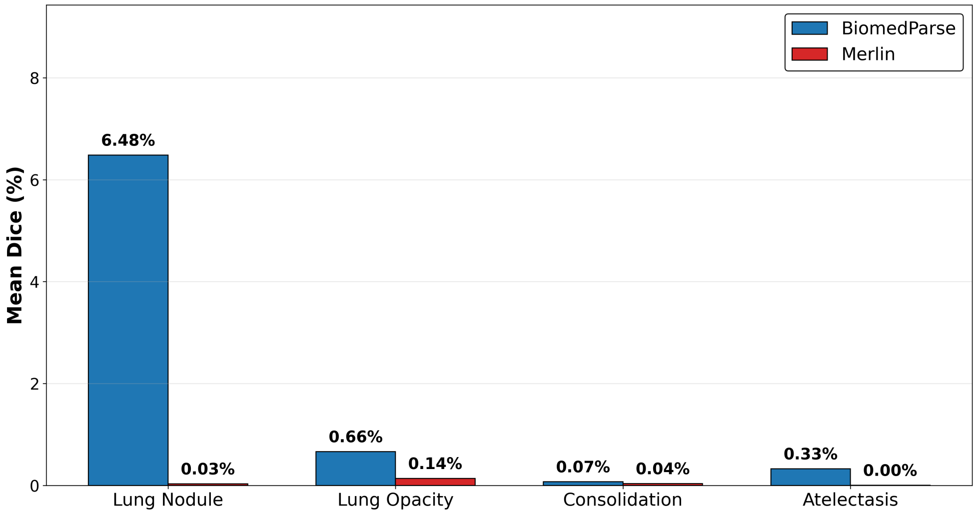

Figure 4.5: Mean Dice comparison by lesion class.

that object-like lesions with clearer boundaries are easier for text-guided segmentation models to localize. In contrast, diffuse abnormalities often have ambiguous borders and variable appearance, making exact spatial overlap more difficult.

4.4.2 Qualitative Comparison of Localization Outcomes

The qualitative BiomedParse example in Figure 4.6 further illustrates the gap between abnormality detection and precise grounding. In the consolidation example, the predicted region covers a broad area of the lung, while the ground-truth annotation is more localized. This indicates that the model responds to the general lung region associated with the abnormality, but it does not accurately preserve the lesion boundary, extent, or class-specific spatial location.

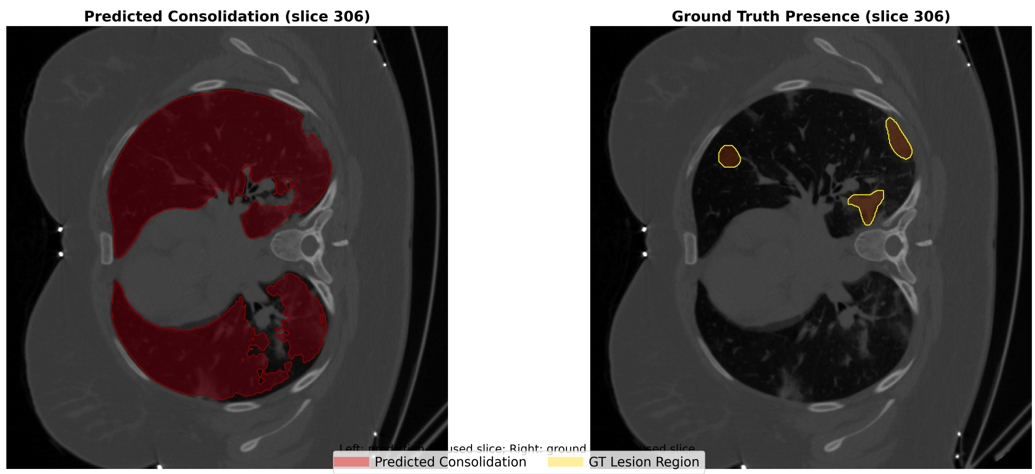

Figure 4.6: Ground-truth and BiomedParse-predicted localization for consolidation.

The LC-KSVD visualization in Figure 4.7 shows a related limitation. The prediction map highlights a suspicious region in the lung, indicating some sensitivity to abnormal tissue patterns. However, the activated region is not fully aligned with the ground-truth lesion location. This suggests that the sparse representation captures some abnormal visual structure, but further refinement is needed to improve anatomical alignment.

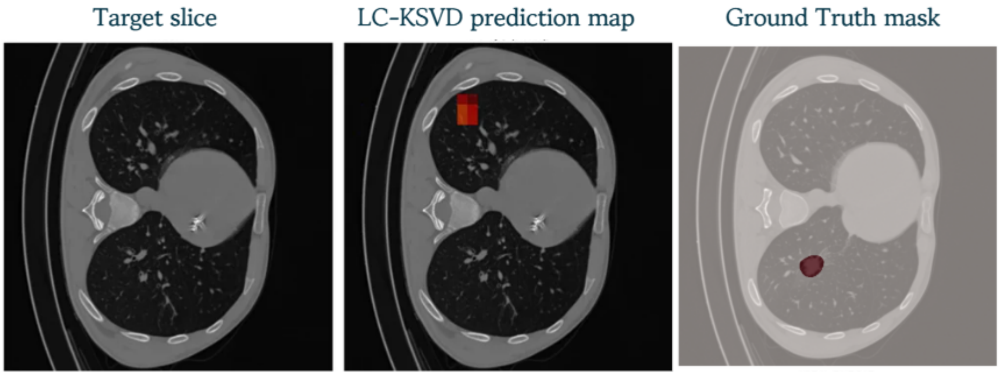

Figure 4.7: Ground-truth and LC-KSVD-predicted localization for consolidation.

Overall, the localization results show that both foundation models and the proposed sparse representation approach face challenges in producing precise lesion-level grounding in 3D CT. This supports the need for evaluation beyond classification scores alone.

4.5 Inherent Explainability Evaluation

Axis C focuses on whether model predictions are supported by clinically meaningful evidence. This is a key motivation for the proposed LC-KSVD approach. Deep learning models such as Merlin, CT-CLIP, and BiomedParse can be evaluated with post-hoc explanation methods or grounding maps, but these explanations are not always part of the model's internal decision process. LC-KSVD, in contrast, provides a more direct form of traceability through its dictionary atoms and sparse coefficients.

4.5.1 Atom-to-Voxel Traceability

LC-KSVD represents input patches using a learned dictionary and sparse activation coefficients. Since the reconstruction and classification processes are based on linear combinations of dictionary atoms, the contribution of selected atoms can be traced back to the corresponding voxel regions in the CT volume. This provides a direct connection between the model's prediction and the spatial image patterns that contributed to it.

This atom-to-voxel traceability is useful because it allows the model's decision to be inspected in terms of activated dictionary components. Instead of relying only on a final disease score, the user can examine which atoms were activated and where their contributions appear in the volume.

4.5.2 Spatial Faithfulness of Explanations

The explanation quality is evaluated using attribution maps generated from model activations or atom contributions. These maps are compared with the ground-truth lesion masks using the following metrics:

- **Attribution-Mask IoU:** Measures how much the explanation map overlaps with the true lesion region.
- **Pointing Game Accuracy:** Checks whether the strongest explanation point falls inside the annotated lesion region.
- **Energy Inside Mask:** Measures how much of the explanation signal is concentrated inside the ground-truth mask.

Together, these metrics assess whether the explanation is spatially faithful to the disease region. A model with good interpretability should not only predict the correct abnormality, but should also base its decision on the relevant anatomical evidence.

For LC-KSVD, these metrics are particularly meaningful because the explanation comes from the model structure itself through sparse codes and dictionary atoms. Even when its classification performance is lower than that of a large foundation model, its explanations can provide additional value by making the decision process more transparent and easier to inspect.

4.6 Summary of Evaluation Findings

The evaluation framework provides a multi-dimensional comparison of model performance. Axis A shows that Merlin is the strongest classification baseline, while LC-KSVD demonstrates promising performance for lung opacity, consolidation, and atelectasis. Axis B shows that localization remains difficult for all evaluated models, especially for diffuse abnormalities. BiomedParse performs relatively better on focal lung nodules, but its localization performance is inconsistent across classes. Axis C highlights the main advantage of LC-KSVD: its ability to provide interpretable atom-based explanations that can be traced back to voxel-level regions.

Overall, the results indicate that classification accuracy alone is not sufficient for evaluating medical AI models in 3D chest CT analysis. A reliable model should be accurate, spatially grounded, and explainable. The proposed evaluation framework captures these three requirements and provides a fair basis for comparing sparse interpretable models with modern foundation model baselines.

Chapter 5

Discussion

Our project investigated an interpretable sparse dictionary learning framework for abnormality classification and localization in 3D chest CT scans. A key contribution is the frozen dictionary approach, where normal anatomical patterns are learned first and abnormalities are learned separately. This helps isolate pathological features and improves interpretability.

Unlike many state-of-the-art deep learning models that rely on post-hoc explainability methods such as Grad-CAM, the proposed LC-KSVD framework provides inherent explainability. The generated predictions can be directly linked to dictionary atoms and reconstructed voxel-space regions through atom back-projection, allowing more transparent clinical interpretation.

The experimental observations show that the framework can successfully classify and localize pulmonary nodules, Ground Glass Opacity (GGO), and consolidation patterns from 3D chest CT scans while maintaining competitive performance. The proposed objectives of implementing an interpretable dictionary learning pipeline, generating localization maps, and evaluating the framework against existing approaches were successfully achieved.

In real-world clinical applications, this approach could support radiologists by providing both diagnostic predictions and anatomically grounded explanations, improving trust and transparency in AI-assisted diagnosis systems.

However, the proposed method has several limitations. The framework is currently limited to three abnormalities and does not yet provide a fully unified end-to-end classification and segmentation architecture. In addition, although the explainability is stronger, the classification accuracy may still be lower than large-scale deep learning foundation models. Future work will focus on improving performance, expanding abnormality coverage, and integrating more advanced hybrid architectures.

Appendix A

Appendix 1

The Appendices are numbered as A, B,.. and so on. Sections and Subsections of the appendix then would be A.1, A.1.1, A.2 etc.

Appendix B

Appendix 2

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